

✓ Fresh start — Project Vishnu

Network params : 537,033

Water cells : 26,641

LOG_EVERY : 100

SAVE_EVERY : 500

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TRAINING — Project Vishnu · Dueling DDQN · 200×200 IOR Grid

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Starting episode : 1

Total episodes : 10,000

Max fuel/ep : 5,000

Buffer capacity : 100,000

Batch size : 64

Device : cuda

Auto-save every : 500 episodes → Google Drive

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Ep 100 | ϵ =0.988 | AvgR= -6927.4 | Goal%= 37.0% | Loss=68.7774 | θ ≐ 171

Ep 200 | ϵ =0.976 | AvgR= -5886.4 | Goal%= 56.0% | Loss=187.4415 | θ ≐ 328

Ep 300 | ϵ =0.964 | AvgR= -5607.2 | Goal%= 58.0% | Loss=151.1344 | θ ≐ 473

Ep 400 | ϵ =0.953 | AvgR= -2828.2 | Goal%= 88.0% | Loss=258.7302 | θ ≐ 562

Ep 500 | ϵ =0.941 | AvgR= -1727.0 | Goal%= 93.0% | Loss=212.0690 | θ ≐ 624 | Buf=100,000

 Saved at episode 500 → Drive + Local

Ep 600 | ϵ =0.929 | AvgR= -830.9 | Goal%= 98.0% | Loss=278.6958 | θ ≐ 668

Ep 700 | ϵ =0.917 | AvgR= -492.3 | Goal%= 99.0% | Loss=286.3961 | θ ≐ 700

Ep 800 | ϵ =0.905 | AvgR= -15.3 | Goal%=100.0% | Loss=323.6862 | θ ≐ 720

Ep 900 | ϵ =0.893 | AvgR= 210.4 | Goal%=100.0% | Loss=310.5444 | θ ≐ 734

Ep 1,000 | ϵ =0.881 | AvgR= 236.5 | Goal%=100.0% | Loss=323.4906 | θ ≐ 747 |

Buf=100,000

 Saved at episode 1000 → Drive + Local

Ep 1,100 | ϵ =0.869 | AvgR= 96.2 | Goal%=100.0% | Loss=266.0986 | θ ≐ 762

Ep 1,200 | ϵ =0.858 | AvgR= 154.4 | Goal%=100.0% | Loss=271.0044 | θ ≐ 776

Ep 1,300 | ϵ =0.846 | AvgR= 332.5 | Goal%=100.0% | Loss=273.7070 | θ ≐ 787

Ep 1,400 | ϵ =0.834 | AvgR= 305.3 | Goal%=100.0% | Loss=266.6569 | θ ≐ 797

Ep 1,500 | ϵ =0.822 | AvgR= 356.6 | Goal%=100.0% | Loss=247.1237 | θ ≐ 806 |

Buf=100,000

 Saved at episode 1500 → Drive + Local

Ep 1,600 | ϵ =0.810 | AvgR= 163.4 | Goal%=100.0% | Loss=233.9838 | θ ≐ 820

Ep 1,700 | ϵ =0.798 | AvgR= 410.3 | Goal%=100.0% | Loss=222.1034 | θ ≐ 827

Ep 1,800 | ϵ =0.786 | AvgR= 408.7 | Goal%=100.0% | Loss=204.0256 | θ ≐ 834

Ep 1,900 | ϵ =0.774 | AvgR= 399.3 | Goal%=100.0% | Loss=194.7054 | θ ≐ 843

Ep 2,000 | ϵ =0.763 | AvgR= 253.1 | Goal%= 99.0% | Loss=183.4719 | θ ≐ 852 | Buf=100,000

 Saved at episode 2000 → Drive + Local

Ep 2,100 | ϵ =0.751 | AvgR= 417.9 | Goal%=100.0% | Loss=189.0294 | θ ≐ 859

Ep 2,200 | ϵ =0.739 | AvgR= 414.2 | Goal%=100.0% | Loss=196.1788 | θ ≐ 867

Ep 2,300 | ϵ =0.727 | AvgR= 465.2 | Goal%=100.0% | Loss=196.3539 | θ ≐ 873

Ep 2,400 | ϵ =0.715 | AvgR= 328.9 | Goal%= 99.0% | Loss=157.6460 | θ ≐ 882
Ep 2,500 | ϵ =0.703 | AvgR= 431.5 | Goal%=100.0% | Loss=157.6632 | θ ≐ 889 |

Buf=100,000

 Saved at episode 2500 → Drive + Local

Ep 2,600 | ϵ =0.691 | AvgR= 323.9 | Goal%= 99.0% | Loss=139.2530 | θ ≐ 899
Ep 2,700 | ϵ =0.679 | AvgR= 441.7 | Goal%=100.0% | Loss=133.4796 | θ ≐ 906
Ep 2,800 | ϵ =0.668 | AvgR= 435.3 | Goal%=100.0% | Loss=140.7976 | θ ≐ 914
Ep 2,900 | ϵ =0.656 | AvgR= 492.3 | Goal%=100.0% | Loss=129.8397 | θ ≐ 921
Ep 3,000 | ϵ =0.644 | AvgR= 474.4 | Goal%=100.0% | Loss=136.5555 | θ ≐ 927 |

Buf=100,000

 Saved at episode 3000 → Drive + Local

Ep 3,100 | ϵ =0.632 | AvgR= 479.1 | Goal%=100.0% | Loss=117.2180 | θ ≐ 934
Ep 3,200 | ϵ =0.620 | AvgR= 466.9 | Goal%=100.0% | Loss=133.2013 | θ ≐ 941
Ep 3,300 | ϵ =0.608 | AvgR= 361.1 | Goal%= 99.0% | Loss=137.7502 | θ ≐ 950
Ep 3,400 | ϵ =0.596 | AvgR= 325.6 | Goal%= 99.0% | Loss=131.0064 | θ ≐ 961
Ep 3,500 | ϵ =0.584 | AvgR= 469.7 | Goal%=100.0% | Loss=122.3896 | θ ≐ 968 |

Buf=100,000

 Saved at episode 3500 → Drive + Local

Ep 3,600 | ϵ =0.573 | AvgR= 506.9 | Goal%=100.0% | Loss=138.2424 | θ ≐ 975
Ep 3,700 | ϵ =0.561 | AvgR= 491.7 | Goal%=100.0% | Loss=133.4368 | θ ≐ 982
Ep 3,800 | ϵ =0.549 | AvgR= 489.6 | Goal%=100.0% | Loss=131.7513 | θ ≐ 990
Ep 3,900 | ϵ =0.537 | AvgR= 437.4 | Goal%=100.0% | Loss=129.4965 | θ ≐ 998
Ep 4,000 | ϵ =0.525 | AvgR= 434.6 | Goal%=100.0% | Loss=142.1973 | θ ≐ 1007 |

Buf=100,000

 Saved at episode 4000 → Drive + Local

Ep 4,100 | ϵ =0.513 | AvgR= 466.8 | Goal%=100.0% | Loss=140.1829 | θ ≐ 1016
Ep 4,200 | ϵ =0.501 | AvgR= 412.0 | Goal%=100.0% | Loss=163.1849 | θ ≐ 1025
Ep 4,300 | ϵ =0.489 | AvgR= 521.3 | Goal%=100.0% | Loss=163.7470 | θ ≐ 1032
Ep 4,400 | ϵ =0.478 | AvgR= 524.5 | Goal%=100.0% | Loss=136.7689 | θ ≐ 1040
Ep 4,500 | ϵ =0.466 | AvgR= 394.5 | Goal%= 99.0% | Loss=116.6077 | θ ≐ 1050 |

Buf=100,000

 Saved at episode 4500 → Drive + Local

Ep 4,600 | ϵ =0.454 | AvgR= 478.6 | Goal%=100.0% | Loss=131.4759 | θ ≐ 1058
Ep 4,700 | ϵ =0.442 | AvgR= 243.5 | Goal%= 99.0% | Loss=130.6321 | θ ≐ 1072
Ep 4,800 | ϵ =0.430 | AvgR= 224.2 | Goal%=100.0% | Loss=116.5896 | θ ≐ 1087
Ep 4,900 | ϵ =0.418 | AvgR= 249.4 | Goal%= 99.0% | Loss=119.4127 | θ ≐ 1101
Ep 5,000 | ϵ =0.406 | AvgR= 145.2 | Goal%= 98.0% | Loss=134.3298 | θ ≐ 1117 |

Buf=100,000

 Saved at episode 5000 → Drive + Local

Ep 5,100 | ϵ =0.394 | AvgR= 144.9 | Goal%= 99.0% | Loss=169.4876 | θ ≐ 1131
Ep 5,200 | ϵ =0.383 | AvgR= 85.8 | Goal%= 99.0% | Loss=150.8329 | θ ≐ 1150
Ep 5,300 | ϵ =0.371 | AvgR= 188.2 | Goal%= 99.0% | Loss=173.8122 | θ ≐ 1167
Ep 5,400 | ϵ =0.359 | AvgR= -383.2 | Goal%= 97.0% | Loss=126.1988 | θ ≐ 1195
Ep 5,500 | ϵ =0.347 | AvgR= -334.2 | Goal%= 99.0% | Loss=123.2713 | θ ≐ 1225 |

Buf=100,000

 Saved at episode 5500 → Drive + Local

Ep 5,600 | $\epsilon=0.335$ | AvgR= -594.9 | Goal%= 96.0% | Loss=130.2311 | $\theta \approx 1259$
Ep 5,700 | $\epsilon=0.323$ | AvgR= -5397.7 | Goal%= 59.0% | Loss=34.3288 | $\theta \approx 1401$
Ep 5,800 | $\epsilon=0.311$ | AvgR= -9542.2 | Goal%= 19.0% | Loss=16.6994 | $\theta \approx 1623$
Ep 5,900 | $\epsilon=0.299$ | AvgR= -10086.7 | Goal%= 13.0% | Loss=11.0529 | $\theta \approx 1848$
Ep 6,000 | $\epsilon=0.288$ | AvgR= -9182.2 | Goal%= 22.0% | Loss=17.5317 | $\theta \approx 2062$ | Buf=100,000

 Saved at episode 6000 → Drive + Local

Ep 6,100 | $\epsilon=0.276$ | AvgR= -10664.9 | Goal%= 5.0% | Loss=9.1873 | $\theta \approx 2302$
Ep 6,200 | $\epsilon=0.264$ | AvgR= -10371.5 | Goal%= 9.0% | Loss=14.1319 | $\theta \approx 2537$
Ep 6,300 | $\epsilon=0.252$ | AvgR= -11219.5 | Goal%= 8.0% | Loss=15.7320 | $\theta \approx 2777$

Ep 7,100 | $\epsilon=0.157$ | AvgR= -11092.5 | Goal%= 3.0% | Loss=5.4828 | $\theta \approx 4686$
Ep 7,200 | $\epsilon=0.145$ | AvgR= -10986.7 | Goal%= 3.0% | Loss=11.9321 | $\theta \approx 4933$
Ep 7,300 | $\epsilon=0.133$ | AvgR= -10797.6 | Goal%= 8.0% | Loss=13.6535 | $\theta \approx 5168$
Ep 7,400 | $\epsilon=0.121$ | AvgR= -11055.2 | Goal%= 2.0% | Loss=4.4393 | $\theta \approx 5413$
Ep 7,500 | $\epsilon=0.109$ | AvgR= -10659.3 | Goal%= 6.0% | Loss=10.4412 | $\theta \approx 5652$ | Buf=100,000

 Saved at episode 7500 → Drive + Local

Ep 7,600 | $\epsilon=0.098$ | AvgR= -10940.7 | Goal%= 7.0% | Loss=11.5494 | $\theta \approx 5894$
Ep 7,700 | $\epsilon=0.086$ | AvgR= -11944.4 | Goal%= 5.0% | Loss=9.9484 | $\theta \approx 6141$
Ep 7,800 | $\epsilon=0.074$ | AvgR= -9846.6 | Goal%= 10.0% | Loss=11.1627 | $\theta \approx 6373$
Ep 7,900 | $\epsilon=0.062$ | AvgR= -12383.0 | Goal%= 3.0% | Loss=10.0755 | $\theta \approx 6619$
Ep 8,000 | $\epsilon=0.050$ | AvgR= -11236.6 | Goal%= 6.0% | Loss=7.6498 | $\theta \approx 6859$ | Buf=100,000

 Saved at episode 8000 → Drive + Local

Ep 8,100 | $\epsilon=0.050$ | AvgR= -10502.0 | Goal%= 6.0% | Loss=7.0747 | $\theta \approx 7102$
Ep 8,200 | $\epsilon=0.050$ | AvgR= -10806.7 | Goal%= 9.0% | Loss=4.5368 | $\theta \approx 7341$
Ep 8,300 | $\epsilon=0.050$ | AvgR= -9886.6 | Goal%= 10.0% | Loss=7.0804 | $\theta \approx 7574$
Ep 8,400 | $\epsilon=0.050$ | AvgR= -10098.7 | Goal%= 3.0% | Loss=14.3454 | $\theta \approx 7818$
Ep 8,500 | $\epsilon=0.050$ | AvgR= -9941.5 | Goal%= 8.0% | Loss=8.8552 | $\theta \approx 8057$ | Buf=100,000

 Saved at episode 8500 → Drive + Local

Ep 8,600 | $\epsilon=0.050$ | AvgR= -11332.9 | Goal%= 10.0% | Loss=6.8289 | $\theta \approx 8292$
Ep 8,700 | $\epsilon=0.050$ | AvgR= -13630.8 | Goal%= 2.0% | Loss=6.8581 | $\theta \approx 8539$
Ep 8,800 | $\epsilon=0.050$ | AvgR= -11379.4 | Goal%= 1.0% | Loss=14.5429 | $\theta \approx 8787$
Ep 8,900 | $\epsilon=0.050$ | AvgR= -10364.5 | Goal%= 8.0% | Loss=7.3806 | $\theta \approx 9023$
Ep 9,000 | $\epsilon=0.050$ | AvgR= -9705.4 | Goal%= 12.0% | Loss=7.5218 | $\theta \approx 9252$ | Buf=100,000

 Saved at episode 9000 → Drive + Local

Ep 9,100 | $\epsilon=0.050$ | AvgR= -10180.2 | Goal%= 7.0% | Loss=8.1767 | $\theta \approx 9493$
Ep 9,200 | $\epsilon=0.050$ | AvgR= -9318.0 | Goal%= 17.0% | Loss=7.3080 | $\theta \approx 9715$
Ep 9,300 | $\epsilon=0.050$ | AvgR= -8529.1 | Goal%= 28.0% | Loss=11.6148 | $\theta \approx 9916$
Ep 9,400 | $\epsilon=0.050$ | AvgR= -10401.5 | Goal%= 5.0% | Loss=4.4206 | $\theta \approx 10158$
Ep 9,500 | $\epsilon=0.050$ | AvgR= -9680.8 | Goal%= 7.0% | Loss=6.0546 | $\theta \approx 10396$ | Buf=100,000

 Saved at episode 9500 → Drive + Local

Ep 9,600 | $\epsilon=0.050$ | AvgR= -9714.5 | Goal%= 9.0% | Loss=6.4971 | $\theta \approx 10630$
Ep 9,700 | $\epsilon=0.050$ | AvgR= -9373.4 | Goal%= 8.0% | Loss=7.3711 | $\theta \approx 10863$
Ep 9,800 | $\epsilon=0.050$ | AvgR= -9363.5 | Goal%= 7.0% | Loss=3.9140 | $\theta \approx 11102$
Ep 9,900 | $\epsilon=0.050$ | AvgR= -11427.9 | Goal%= 5.0% | Loss=10.5440 | $\theta \approx 11344$

Ep 10,000 | $\epsilon=0.050$ | AvgR= -9890.2 | Goal%= 11.0% | Loss=4.5906 | $\theta=11577$ |
Buf=100,000

 Saved at episode 10000 → Drive + Local

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✓ Training complete! Total episodes: 10,000

Final ϵ : 0.0500

Final goal rate : 11.0%

Target updates : 11577

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 Saved at episode 10000 → Drive + Local

Results:

